

ECO-VIBE: APPLICATION FOR BUYING AND SELLING SECONDHAND CLOTHES

Carol Stefanya Velasco Rodríguez, David Eduardo Muñoz Mariño
Universidad Distrital Francisco Jose de Caldas
Faculty of Engineering - Systems Engineering
Bogotá D.C., February 2025

CONTENTS

I	Introduction	1
II	User Stories	1
II-A	Questions	1
II-B	User stories	1
III	Functional and Non-Functional Requirements	2
III-A	Functional Requirements	2
III-B	Non-Functional Requirements	2
IV	UML Diagrams	2
IV-A	Deployment diagram	2
IV-B	Sequence diagram	2
IV-C	Activity diagram	2
V	Conceptual Design	3
V-A	Web GUI Mockups	3
V-B	CRC cards	4
VI	System Architecture	4
VI-A	Class diagram	4
VII	Backend definition	4
VII-A	Java Backend (Spring Boot)	4
VII-B	Python Backend (Flask)	4
VII-C	Connection Between Java Backend and Python Backend	5
VIII	SOLID Principle	5
VIII-A	Single Responsibility Principle (SRP)	5
VIII-B	Open/Closed Principle (OCP)	5
VIII-C	Liskov Substitution Principle (LSP)	5
VIII-D	Interface Segregation Principle (ISP)	5
VIII-E	Dependency Inversion Principle (DIP)	5
IX	Conclusions	5

LIST OF FIGURES

1	User interest in second-hand clothing applications	1
2	How users prefer to view products in the app	1
3	User preferences for product details	1
4	Deployment diagram	2
5	Sequence diagram	2
6	Activity diagram	2
7	Login and Sign Up	3
8	Upload Product	3
9	Search Product	3
10	Buy Product	3
11	Rate Product	3
12	CRC cards	4
13	Class diagram	4

ECO-VIBE: APPLICATION FOR BUYING AND SELLING SECONDHAND CLOTHES

Abstract—ECO-VIBE is an e-commerce platform developed to support the buying and selling of secondhand clothing. The system integrates Java and Python web services to manage product listings, search functions, and user interactions. Its design focuses on efficiency, ensuring smooth operations across different components. This report details the platform’s architecture, backend development, and key processes, explaining how various technologies work together to enhance functionality and scalability.

I. INTRODUCTION

The rise of digital platforms has changed how people buy and sell secondhand clothing. ECO-VIBE aims to create an easy-to-use and efficient online marketplace where users can quickly buy and sell clothes.

This project focuses not only on user experience but also on integrating web services developed in Java and Python to manage the platform’s different features. This architecture improves product listings, searching, and purchasing.

This report explains the development, design, and implementation of the application, covering both business logic and the connection between backend services.

II. USER STORIES

A survey was carried out on 20 potential users with the aim of knowing their preferences and needs when using a second-hand clothing sales application. The questions aimed to identify the main goals of the users, the desired features in the application, the way the products are displayed, the necessary information about the products and the importance of direct contact with other users.

A. Questions

The questions and their results were:

1. What is your main interest when using a virtual second-hand clothing application?

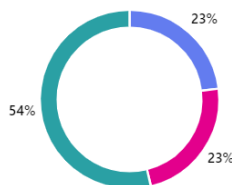


Fig. 1. User interest in second-hand clothing applications

Options:

- Buy second-hand clothes (Blue)
- Sell second-hand clothes (Pink)
- Both options (Green)

2. How would you like the products to be displayed in the app?

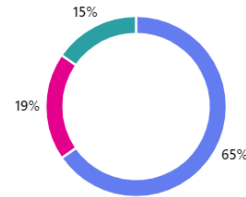


Fig. 2. How users prefer to view products in the app

Options:

- With photos or videos (Blue)
- With opinions and reviews from other users (Pink)
- Others: (Green)

- Both

- All of the above

3. What information about the product would you like to see provided in the app?

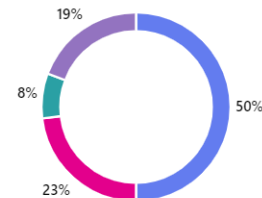


Fig. 3. User preferences for product details

Options:

- Detailed product description (Blue)
- Product condition (new, used, etc) (Pink)
- Size and measurements (Green)
- Others: (Purple)
- All of the above

B. User stories

Once the responses are collected, user stories are created:

- "As a seller, I want to be able to add detailed descriptions and high-quality photographs to show my products in the best way possible"

- "As a user I want to be able to see the products with photos and detailed descriptions to have a clear idea of what I am buying"

- "As a buyer, I want to be able to see detailed information about the product, such as condition and measurements, to make an informed decision"

After analyzing user responses, it was observed that most users value the ability to view products with photographs and detailed descriptions. The importance of detailed product information, such as conditions and measurements, was also highlighted. The proposed functionalities will cover the needs of users, and additional suggestions will be taken into account to improve the user experience in the second-hand clothing sales application.

III. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

A. Functional Requirements

- User Registration: Users create an account by providing a username and password.
- Product Listing: Sellers upload images, descriptions, and prices of their secondhand clothing items.
- Product Search: Buyers search for products using keywords.
- Product Purchase: To buy a product, users enter their address, bank details, and phone number.
- Reviews and Ratings: Buyers leave a review of both the product and the seller after completing a purchase.

B. Non-Functional Requirements

- Usability: The platform is intuitive and easy to navigate.
- Visual Design: The interface has pleasant colors and a clear layout.
- Maintainability: The system allows future updates without affecting functionality.

IV. UML DIAGRAMS

A. Deployment diagram

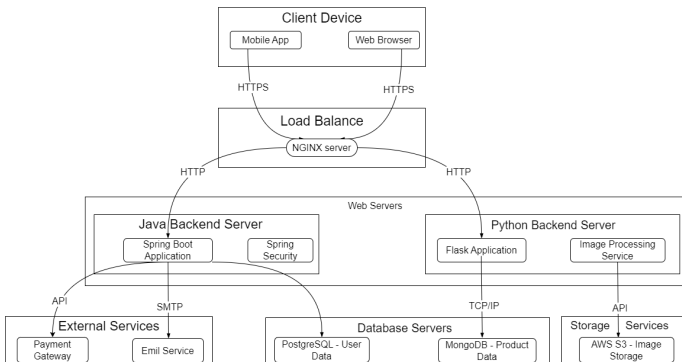


Fig. 4. Deployment diagram

B. Sequence diagram

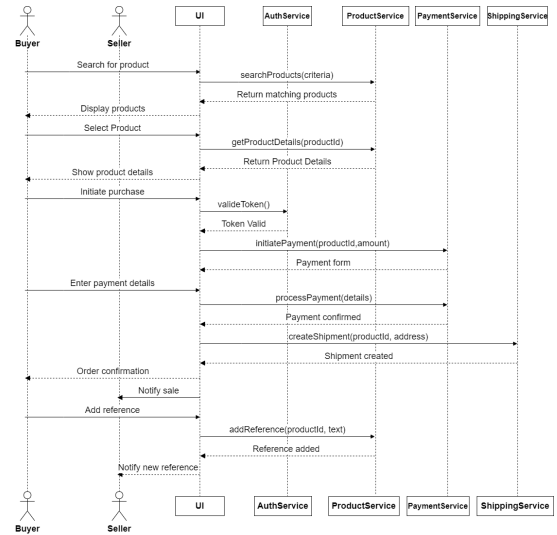


Fig. 5. Sequence diagram

C. Activity diagram

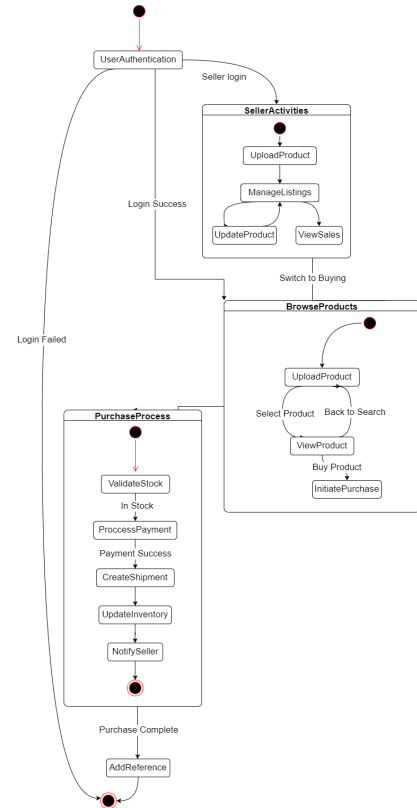


Fig. 6. Activity diagram

V. CONCEPTUAL DESIGN

A. Web GUI Mockups

The design of EcoVibe is based on earth tones and brown colors to create a natural and eco-friendly feel, matching the idea of giving clothes a second life. A minimalist style was chosen to keep the platform simple, with a clean and organized look.

Each mockup is designed to make navigation clear and easy, so users don't feel lost. The elements are arranged in a way that helps users complete actions like searching for products, signing up, or making a purchase. There is also an option to rate the seller's service, allowing users to share their experience and help improve the platform. The font style and button placement were carefully selected to enhance readability and accessibility. Additionally, spacing and contrast ensure that the interface remains visually appealing while being functional for all users.

The Mockups are:

- Mockup - Login and Sign Up

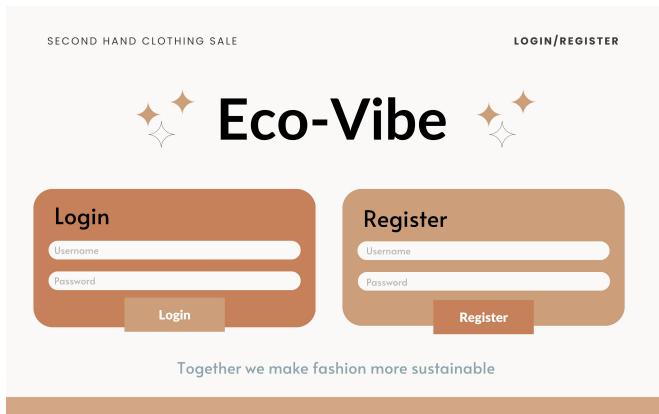


Fig. 7. Login and Sign Up

- Mockup - Upload Product

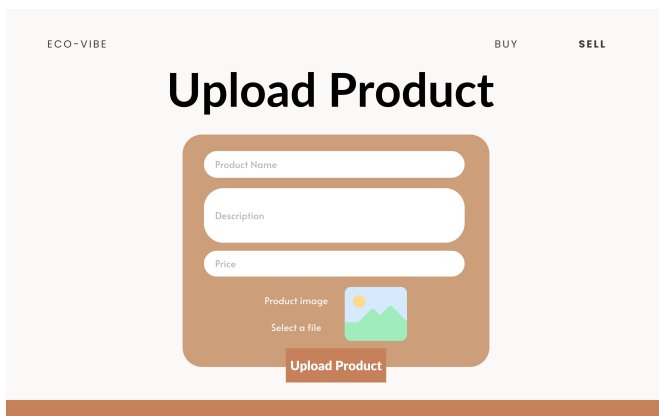


Fig. 8. Upload Product

- Mockup - Search Product

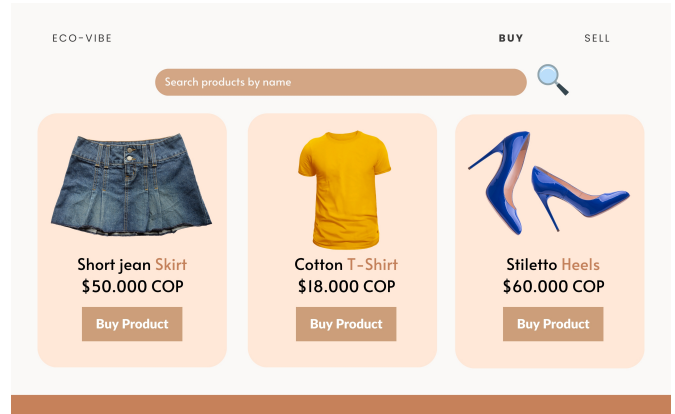


Fig. 9. Search Product

- Mockup - Buy Product



Fig. 10. Buy Product

- Mockup - Rate Product



Fig. 11. Rate Product

B. CRC cards

The CRC cards show the structure and organization of the main components of the EcoVibe application, detailing their responsibilities and interactions. They identify key elements such as users, products, orders, and payments, along with their functions in the system. This representation helps to understand how the different parts of the application work together, making development and maintenance easier.

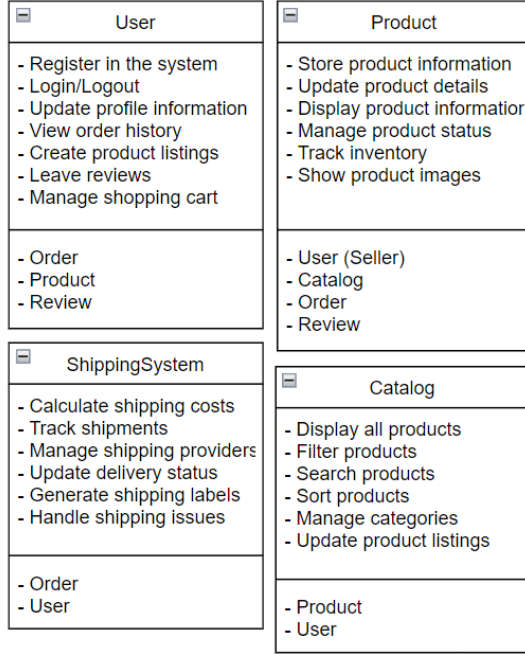


Fig. 12. CRC cards

VI. SYSTEM ARCHITECTURE

A. Class diagram

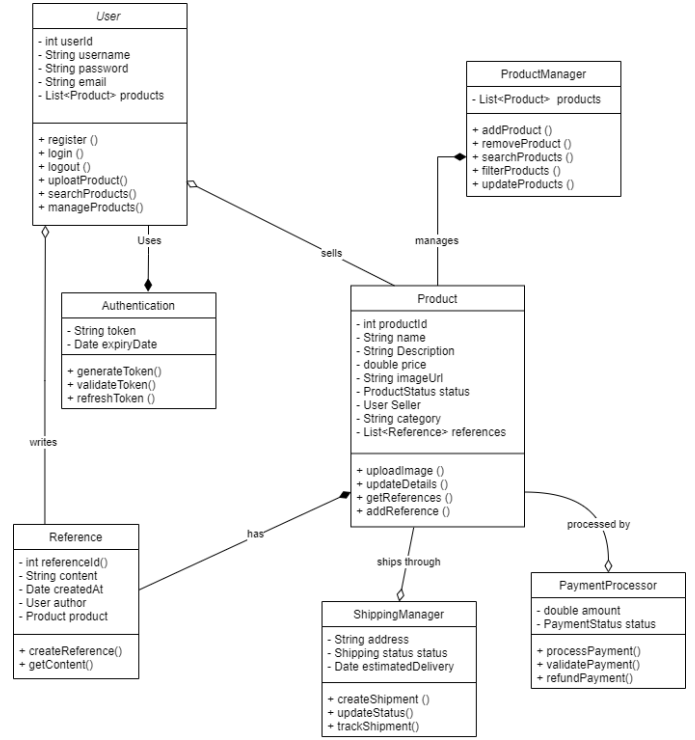


Fig. 13. Class diagram

VII. BACKEND DEFINITION

A. Java Backend (Spring Boot)

• Technologies and Libraries Used:

Spring Boot is the main framework for backend development. Spring Security implements authentication and authorization using JWT, while Spring Web facilitates the creation of REST APIs.

• Web Services and Endpoints:

Authentication (AuthController.java)

The authentication service includes the endpoint POST /api/auth/login, which receives username and password, returning a JWT token if the credentials are correct. It also has GET /api/auth/user/me, a protected endpoint for verifying authentication.

User Registration (RegistrationController.java)

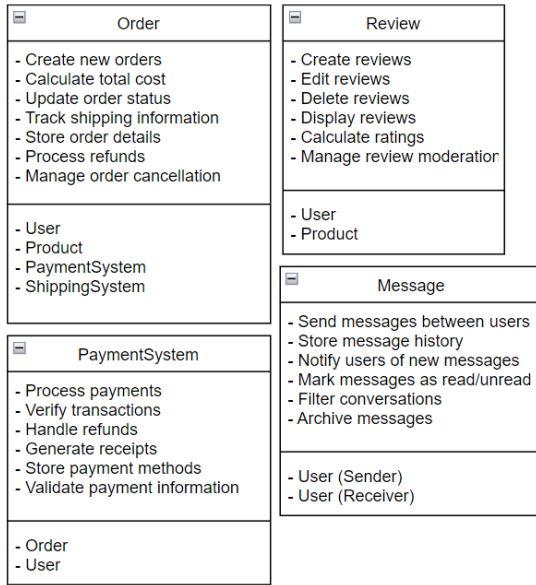
The user registration service provides POST /api/auth/register, which receives username and password, allowing the creation of a new user.

B. Python Backend (Flask)

• Technologies and Libraries Used:

The Python backend uses Flask as a lightweight framework for creating REST APIs. Flask-CORS allows requests from the frontend, and Werkzeug handles file uploads.

• Web Services and Endpoints:



a) *General:*

: • *GET /* Main page.

b) *File Management:*

: • *GET /uploads/filename* Retrieves uploaded files.

c) *Product Management:*

: • *GET /products* Lists all products.

• *POST /products* Creates a new product.

• *GET /products/int:productid* Retrieves details of a specific product.

d) *Product References:*

: • *GET /products/int:productid/references* Lists references for a product.

• *POST /products/int:productid/references* Adds a new reference to a product.

C. Connection Between Java Backend and Python Backend

The Spring Boot backend handles authentication, and Flask may require JWT authentication for certain endpoints. Interaction between both can be achieved by validating tokens generated by Spring Boot in Flask to ensure secure requests.

VIII. SOLID PRINCIPLE

A. Single Responsibility Principle (SRP)

Each class and module has only one responsibility. In Java, *AuthController* handles authentication, while *JwtService* manages token generation and validation. In Python, routes are organized to handle different system functions clearly.

B. Open/Closed Principle (OCP)

The code is designed to allow extensions without modifying its core structure. In Java, *JwtService* can change authentication methods without affecting the rest of the system. In Python, the modular route structure makes it easy to add new features.

C. Liskov Substitution Principle (LSP)

The system architecture allows class replacements without breaking functionality. In Java, interfaces like *UserDetailsService* allow swapping implementations without modifying dependent code.

D. Interface Segregation Principle (ISP)

Interfaces and modules are designed to contain only necessary methods, avoiding unnecessary complexity. In Java, each controller handles only its specific operations, ensuring clean code.

E. Dependency Inversion Principle (DIP)

Dependency injection is used to decouple modules, making the system flexible and scalable. In Java, Spring Boot implements dependency injection with *@Autowired* and *@Service*, allowing easy replacement of components without changing the system structure.

IX. CONCLUSIONS

- *Structure and Organization:* The combination of Java (Spring Boot) and Python (Flask) creates a functional backend, but Flask should be more modular to improve maintenance and scalability.

- *Application of SOLID:* Java follows SOLID principles with a well-defined architecture. Python is functional but needs better separation of responsibilities and dependency management.

- *Importance of User Stories:* Defining user stories was key to understanding client needs and designing features that meet expectations, ensuring a user-focused product.

- *Use of Diagrams:* Diagrams such as architecture, data flow, and use case diagrams helped visualize the system, improve development, and enhance team communication.

- *Mockup Design:* Creating mockups helped define a user-friendly interface before development, improving user experience and reducing later modifications.

- *Scalability and Future Improvements:* It is recommended to refine modularity, improve system documentation, and optimize backend integration for future expansions.